

Advanced Programming

Assignment Submission | Spring 2026

Name	Gaurav Nath
Roll No.	CSB24073
Subject	Advanced Programming
Assignment	Assignment No. 19
Semester	Spring 2026
Framework	React Native (Expo)
Language	JavaScript (JSX)
Device	iPhone 15 via Expo Go

Simple Digital Counter & Theme Toggle App

A single-screen mobile application built with React Native (Expo) that functions as a digital counter with Light Mode and Dark Mode theme toggling. The app demonstrates Flexbox layout, useState hook, dynamic styling, and component-based architecture.

Question

Build a single-screen mobile application using React Native. The app functions as a digital counter that allows users to increment, decrement, and reset a number displayed on the screen. It must also include a Theme Toggle button that switches between Light Mode and Dark Mode.

Implementation Rules:

- Use View, Text, and TouchableOpacity components with Flexbox layout
- Use useState hook to manage counter value and theme mode
- Counter starts at 0 — decrement never goes below 0
- Light Mode: white background, dark text — Dark Mode: dark background, white text
- Theme toggle swaps styles instantly across the entire screen

Must demonstrate:

1. UI Layout using Flexbox with flex, justifyContent, alignItems, fontSize, padding
2. Counter state with useState — increment, decrement with validation, reset
3. Dynamic theme toggling using ternary operators in style objects
4. Clean code with meaningful names: handleIncrement, handleDecrement, toggleTheme

Setup and Running the App

Platform: iPhone 15 + Windows laptop using Expo Go (no Android Studio needed)

Terminal Commands — Setup and Run

```
# Step 1: Install Node.js from https://nodejs.org (LTS version)

# Step 2: Install Expo CLI
npm install -g expo-cli

# Step 3: Create the project
cd "D:\Downloads D Folder\Advanced Programming 2"
npx create-expo-app Assignment19
cd Assignment19

# Step 4: Replace app/index.tsx with the provided code

# Step 5: Start Expo
npx expo start

# Step 6: Scan QR code with iPhone Camera → tap Expo Go link
```

Note: iPhone and laptop must be on the same WiFi network. Expo Go app must be installed from the App Store.

Code — Key Snippets

1. Imports

index.tsx — Imports

```
import React, { useState } from 'react';
import {
  View,
  Text,
  TouchableOpacity,
  StyleSheet,
  StatusBar,
  SafeAreaView,
} from 'react-native';
```

2. State Management — useState Hook

index.tsx — State

```
// Two pieces of state as required:
const [count, setCount] = useState(0); // counter value
const [isDarkMode, setIsDarkMode] = useState(false); // theme mode
```

3. Counter Logic and Theme Toggle Functions

index.tsx — Handler Functions

```
// Increment — adds 1
```

```

const handleIncrement = () => setCount(prev => prev + 1);

// Decrement - never goes below 0 (constraint check)
const handleDecrement = () => {
  if (count > 0) setCount(prev => prev - 1);
};

// Reset - back to 0
const handleReset = () => setCount(0);

// Theme toggle - flips boolean
const toggleTheme = () => setIsDarkMode(prev => !prev);

```

4. Dynamic Theme Object — Ternary Styling

index.tsx — Theme Object

```

const theme = {
  container: isDarkMode ? '#1a1a1a' : '#ffffff',
  text: isDarkMode ? '#ffffff' : '#1a1a1a',
  subText: isDarkMode ? '#aaaaaa' : '#555555',
  card: isDarkMode ? '#2c2c2c' : '#f2f2f2',
  btnInc: '#4CAF50', // green - always
  btnDec: '#f44336', // red - always
  btnReset: '#2196F3', // blue - always
  btnTheme: isDarkMode ? '#FFD700' : '#333333',
  btnThemeTxt: isDarkMode ? '#1a1a1a' : '#ffffff',
  toggleLabel: isDarkMode ? 'Switch to Light Mode ☐' : 'Switch to Dark Mode ☐',
};

```

5. JSX Layout — Component Structure

index.tsx — Return / JSX

```

return (
  <SafeAreaView style={[styles.safe, { backgroundColor: theme.container }]}>
    <StatusBar barStyle={isDarkMode ? 'light-content' : 'dark-content'} />

    <View style={[styles.container, { backgroundColor: theme.container }]}>

      {/* App Heading */}
      <Text style={[styles.title, { color: theme.text }]}>Mindful Counter</Text>
      <Text style={[styles.subtitle, { color: theme.subText }]}>
        Stay focused. Count habits, people, items, or daily goals.
      </Text>

      {/* Info Card */}
      <View style={[styles.infoCard, { backgroundColor: theme.card }]}>
        <Text style={[styles.infoText, { color: theme.subText }]}>
          Your daily minimalist companion for keeping track of repetitions...
        </Text>
      </View>

      {/* Counter Display */}
      <View style={[styles.card, { backgroundColor: theme.card }]}>
        <Text style={[styles.counterValue, { color: theme.text }]}>{count}</Text>
        <Text style={[styles.counterLabel, { color: theme.subText }]}>Current
      </View>

      <Text>Count</Text>

      {/* Increment + Decrement Row */}

```

```

<View style={styles.row}>
  <TouchableOpacity style={[styles.btn, { backgroundColor: theme.btnInc }]}
    onPress={handleIncrement}>
    <Text style={styles.btnText}>>+ Increment</Text>
  </TouchableOpacity>

  <TouchableOpacity
    style={[styles.btn, { backgroundColor: theme.btnDec },
      count === 0 && styles.btnDisabled]}
    onPress={handleDecrement}
    disabled={count === 0}>
    <Text style={styles.btnText}>>- Decrement</Text>
  </TouchableOpacity>
</View>

{/* Reset Button */}
<TouchableOpacity style={[styles.btnWide, { backgroundColor: theme.btnReset }]}
  onPress={handleReset}>
  <Text style={styles.btnText}>>↺ Reset</Text>
</TouchableOpacity>

{/* Theme Toggle */}
<TouchableOpacity style={[styles.btnWide, { backgroundColor: theme.btnTheme }]}
  onPress={toggleTheme}>
  <Text style={[styles.btnText, { color: theme.btnThemeTxt }]}>
    {theme.toggleLabel}
  </Text>
</TouchableOpacity>

</View>
</SafeAreaView>
);

```

6. StyleSheet

index.tsx — StyleSheet.create

```

const styles = StyleSheet.create({
  safe: { flex: 1 },
  container: { flex: 1, justifyContent: 'center', alignItems: 'center',
    paddingHorizontal: 24 },
  title: { fontSize: 32, fontWeight: 'bold', marginBottom: 6, letterSpacing: 1
},
  subtitle: { fontSize: 14, marginBottom: 20 },
  infoCard: { width: '90%', padding: 18, borderRadius: 16, marginBottom: 30 },
  infoText: { textAlign: 'center', fontSize: 14, lineHeight: 22 },
  card: { width: '80%', paddingVertical: 40, borderRadius: 20,
    alignItems: 'center', marginBottom: 40, elevation: 6 },
  counterValue: { fontSize: 80, fontWeight: 'bold' },
  counterLabel: { fontSize: 14, marginTop: 4 },
  row: { flexDirection: 'row', gap: 12, marginBottom: 12 },
  btn: { flex: 1, paddingVertical: 16, borderRadius: 12, alignItems: 'center'
},
  btnWide: { width: '100%', paddingVertical: 16, borderRadius: 12,
    alignItems: 'center', marginBottom: 12 },
  btnDisabled: { opacity: 0.4 },
  btnText: { color: '#ffffff', fontSize: 16, fontWeight: '600' },
});

```

Features Implemented

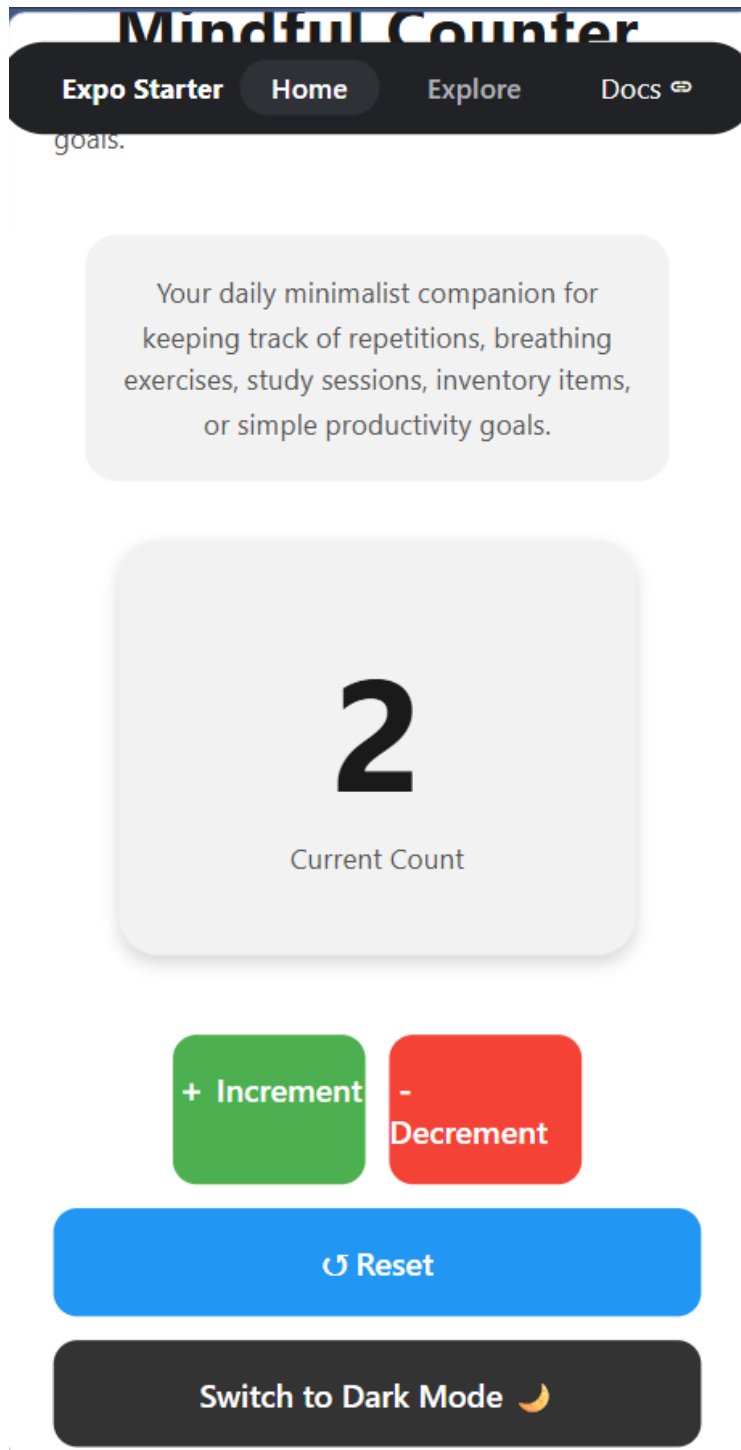
Feature	Implementation	Detail
Counter Display	<code>useState(0)</code>	Shows current count in large bold text on card
Increment	<code>handleIncrement()</code>	Adds 1 to count on button press
Decrement	<code>handleDecrement()</code>	Subtracts 1 — disabled and dimmed at 0
Reset	<code>handleReset()</code>	Sets count back to 0
Theme Toggle	<code>useState(false) + toggleTheme()</code>	Flips <code>isDarkMode</code> boolean on press
Light Mode	<code>isDarkMode = false</code>	White background, dark text
Dark Mode	<code>isDarkMode = true</code>	Dark grey background, white text
Flexbox Layout	<code>flex, justifyContent, alignItems</code>	All elements perfectly centered
Disabled Decrement	<code>disabled={count === 0}</code>	Button dims to 40% opacity at 0
SafeAreaView	SafeAreaView wrapper	Handles iPhone notch and status bar

Concept Summary

- `useState` manages two independent pieces of state — `count` (integer) and `isDarkMode` (boolean) — both trigger a re-render when updated.
- Dynamic styling uses ternary operators inside the theme object so every color switches instantly when `isDarkMode` flips.
- Flexbox centers all UI elements using `flex: 1, justifyContent: 'center', alignItems: 'center'` on the parent container.
- The decrement constraint check (if `count > 0`) ensures the app never crashes or shows negative numbers.
- `TouchableOpacity` provides visual press feedback and the `disabled` prop prevents interaction when count is already 0.

Screenshots

App running on iPhone 15 — Light Mode:



App running on iPhone 15 — Dark Mode:

Expo Starter

Home

Explore

Docs 

Stay focused. Count habits, people, items, or daily goals.

Your daily minimalist companion for keeping track of repetitions, breathing exercises, study sessions, inventory items, or simple productivity goals.

11

Current Count

+ Increment

-
Decrement

 Reset

Switch to Light Mode 